Customer Satisfaction Survey Analysis

Data Cleaning & Power Query Methodology

Timothy Macqueen

Customer Satisfaction Survey Dashboard

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Tools used: Excel, Power Query, Microsoft VS Code Python/Pandas

2. Project Overview

This project analyses customer satisfaction survey data to identify patterns in customer satisfaction, purchasing behaviour, product performance and customer sentiment. Power Query was used to clean, standardise and transform the raw dataset before analysis in Excel PivotTables, PivotCharts and dashboard visualisations.

3.) Source Data

Number of records: 10,000

Number of columns: 13

What each record represents: One customer survey response.

Important fields: RespondentID, Age, Gender, Province, Product, Satisfaction, Recommend, Purchase Amount, Comments.

The dataset used for this portfolio project is a synthetic customer satisfaction dataset created for data-analysis practice. It was generated using Python, Faker and Pandas in Microsoft Visual Studio Code, exported as a .xlsx file (Figure 1) and subsequently processed in Excel Power Query.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	RespondentID	Date	Age	Gender	Province	Product	Satisfaction	Recommend	Purchase Amount	Payment Method	Delivery Type	Loyalty Member	Comments
41	R09146	2025-01-09		Female	North West	Storage Basket	3	Yes	62.22	Debit Card	Click & Collect	No	Excellent value
42	R07774	2025-02-12	29	Male	Western Cape	Samsung Galaxy Tablet	5	Yes	2770.76	Mobile Payment	In Store	No	Excellent product
43	R01569	2024-09-12	65	Male	Western Cape	Facial Cleanser	4	Yes	3378.44	Debit Card	Home Delivery	Yes	Instructions unclear
44	R02733	2025-04-21	44	Other	Free State	Vacuum Cleaner	1	No	3053.06	Cash	Click & Collect	No	Delivery delayed
45	R00170	2024-03-16	46	Male	Free State	Bose Soundbar	3	No	2484.84	Mobile Payment	In Store	No	Excellent value
46	R09284	2024-07-09	59	Male	Western Cape	Laptop Backpack	2	Yes	960.97	Cash	Home Delivery	No	Easy online ordering
47	R00488	2024-02-13	130	Other	NORTHERN CAPE	Samsung Galaxy Tablet	3	No	2236.63	Credit Card	Click & Collect	Yes	Easy to assemble
48	R07916	2025-07-10	71	Other	Free State	Winter Jacket	1	No	405.34	Debit Card	Home Delivery	No	Battery life disappointing
49	R05685	2024-08-03	74	Other	Mpumalanga	Apple AirPods Pro	4	Yes	2180.31	Debit Card	In Store	No	Helpful staff
50	R09981	2025-05-07	37	Male	Northern Cape	Fruit Juice	5	No	3042.34	Credit Card	Home Delivery	No	Fits perfectly
51	R04690	2025-06-02	49		Mpumalanga	Laptop Backpack	5	Yes	770.17	Debit Card	Click & Collect	No	Would definitely buy again
52	R05082	2025-03-27	42	Other	Western Cape	Ceiling Fan	2	Yes	302.98	Mobile Payment	Click & Collect	No	Battery life disappointing
53	R02288	2024-11-24	66	Other	Free State	Camping Chair	4	No	2486.85	EFT	In Store	Yes	Long waiting time
54	R09456	2025-03-21	67	Male		Winter Jacket	1	Yes	1769.72	EFT	Click & Collect	Yes	Product exceeded expectations
55	R06395	2024-09-23	58	Male	KwaZulu-Natal		4	Yes	193.14	Debit Card	In Store	Yes	Very fast delivery
56	R01982	2025-05-12	33	Other	Gauteng	Watch Strap	5	Yes	2781.53	Cash	Home Delivery	Yes	Great customer service
57	R07244	2024-05-18		Other	Gauteng		2	Yes	3479.49	EFT	In Store	No	Product exceeded expectations
58	R05913	2025-02-13	-5	Male	Western Cape	Laundry Hamper	3	Yes	1794.4	Cash	In Store	No	Item missing accessories
59	R04848	2024-08-03	28	Other	Limpopo	Dell Inspiron Laptop	3	No	2649.73	Debit Card	Home Delivery	Yes	Instructions unclear
60	R02887	2025-03-01	26	Female	Mpumalanga	Laundry Hamper	1	Yes		EFT	In Store	No	Average quality
61	R07620	2025-07-03	29	Female	KwaZulu-Natal	Floor Lamp	4	Yes	398.01	EFT	In Store	Yes	Very comfortable
62	R08758	2024-05-25		Male	KwaZulu-Natal	Canon EOS DSLR Camera	5	No	206.5	EFT	Click & Collect	No	Would definitely buy again
63	R09552	2024-02-24	58	Other	Western Cape	Ceiling Fan	5	No	2263.44	Credit Card	Click & Collect	Yes	Easy to assemble

customer_satisfaction_raw

Figure 1: Raw dataset before cleaning

4. Initial Data Quality Assessment

The following is a summary of the data checks made during cleaning in Power Query (Table 1).

Table 1: Data Quality Checks in Power Query

Data Quality Check	Issue Identified	Action
Missing values	Missing values in selected fields	Investigated and handled
Duplicate records	Duplicate records checked	Removed where appropriate
Data types	Inconsistent data types	Corrected
Spelling	Inconsistent category names	Standardised
Province names	Inconsistent entries	Standardised
Dates	Date formatting inconsistencies	Converted to Date
Numeric fields	Numeric values stored incorrectly	Converted to appropriate type
Product names	Inconsistent text formatting	Cleaned

5. Power Query Cleaning Process

See Appendix A for more detailed information of column-by-column edits

Step 1 — Imported the data

The raw customer satisfaction dataset was imported into Excel Power Query to create a repeatable data-cleaning workflow. Became known as Source Data. Unit of observation: Each record represents one customer survey response. Each row contains demographic, purchasing, satisfaction and feedback information for an individual respondent.

Step 2 — Corrected data types

Data was corrected to the correct data type within Power Query (Table 2).

Table 2: Corrected Data types in Power Query

Field	Data Type
Respondent ID	Whole Number/Text
Date	Date
Age	Whole Number

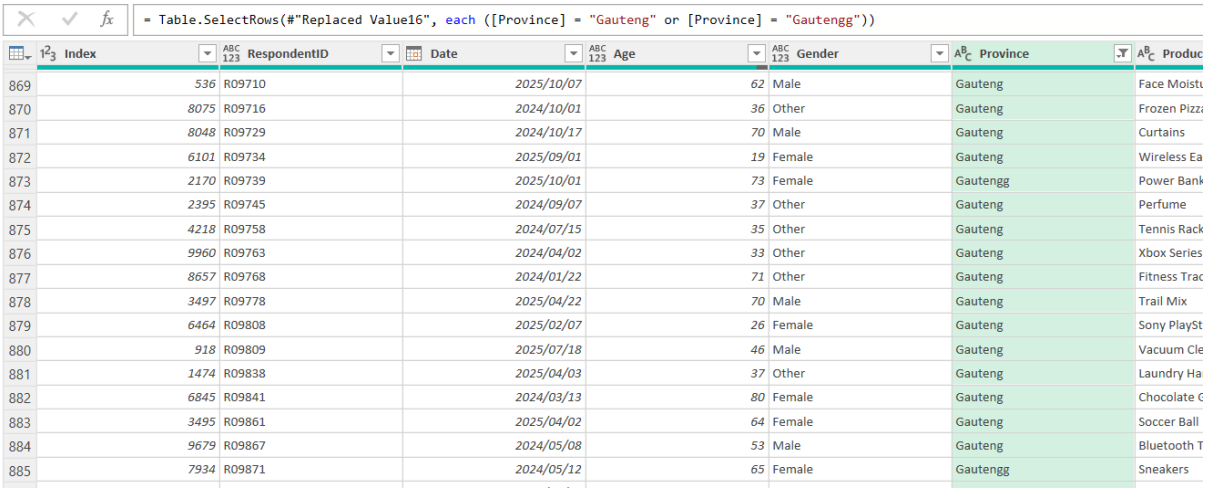
Purchase Amount	Decimal Number
Satisfaction	Whole Number
Province	Text
Department	Text
Product	Text

Step 3 — Cleaned text fields

Text fields were cleaned using the Trim, Clean and Replace Values functions in Power Query to remove unnecessary spaces and standardise category values. Inconsistent categorical values were replaced with the correct standardised values (See Appendix A). This was particularly useful for fields such as Province and Product.

Step 4 — Handled inconsistent categorical values

Categorical values were reviewed for spelling inconsistencies. Incorrect category labels were standardised to ensure that equivalent categories were grouped correctly during PivotTable analysis. For example: Before: Gautengg After: Gauteng (Figure 2).



	Index	RespondentID	Date	Age	Gender	Province	Produc
869	536	R09710	2025/10/07	62	Male	Gauteng	Face Moistu
870	8075	R09716	2024/10/01	36	Other	Gauteng	Frozen Pizz
871	8048	R09729	2024/10/17	70	Male	Gauteng	Curtains
872	6101	R09734	2025/09/01	19	Female	Gauteng	Wireless Ea
873	2170	R09739	2025/10/01	73	Female	Gautengg	Power Bank
874	2395	R09745	2024/09/07	37	Other	Gauteng	Perfume
875	4218	R09758	2024/07/15	35	Other	Gauteng	Tennis Rack
876	9960	R09763	2024/04/02	33	Other	Gauteng	Xbox Series
877	8657	R09768	2024/01/22	71	Other	Gauteng	Fitness Trac
878	3497	R09778	2025/04/22	70	Male	Gauteng	Trail Mix
879	6464	R09808	2025/02/07	26	Female	Gauteng	Sony PlaySt
880	918	R09809	2025/07/18	46	Male	Gauteng	Vacuum Cle
881	1474	R09838	2025/04/03	37	Other	Gauteng	Laundry Ha
882	6845	R09841	2024/03/13	80	Female	Gauteng	Chocolate C
883	3495	R09861	2025/04/02	64	Female	Gauteng	Soccer Ball
884	9679	R09867	2024/05/08	53	Male	Gauteng	Bluetooth T
885	7934	R09871	2024/05/12	65	Female	Gautengg	Sneakers

Figure 2: Standardization of Categorical values in this case Gautengg → Gauteng

Step 5 — Checked missing values

Missing values were identified using Power Query filtering and column profiling (Figures 3 and 4). Fields required for the analysis were reviewed and appropriate actions were taken where missing values were identified.

Step 6 — Checked duplicates

Duplicate records were checked using the relevant respondent/record identifiers. Duplicate records were removed where they represented obvious duplicate observations. This was especially the case for the RespondentID column.

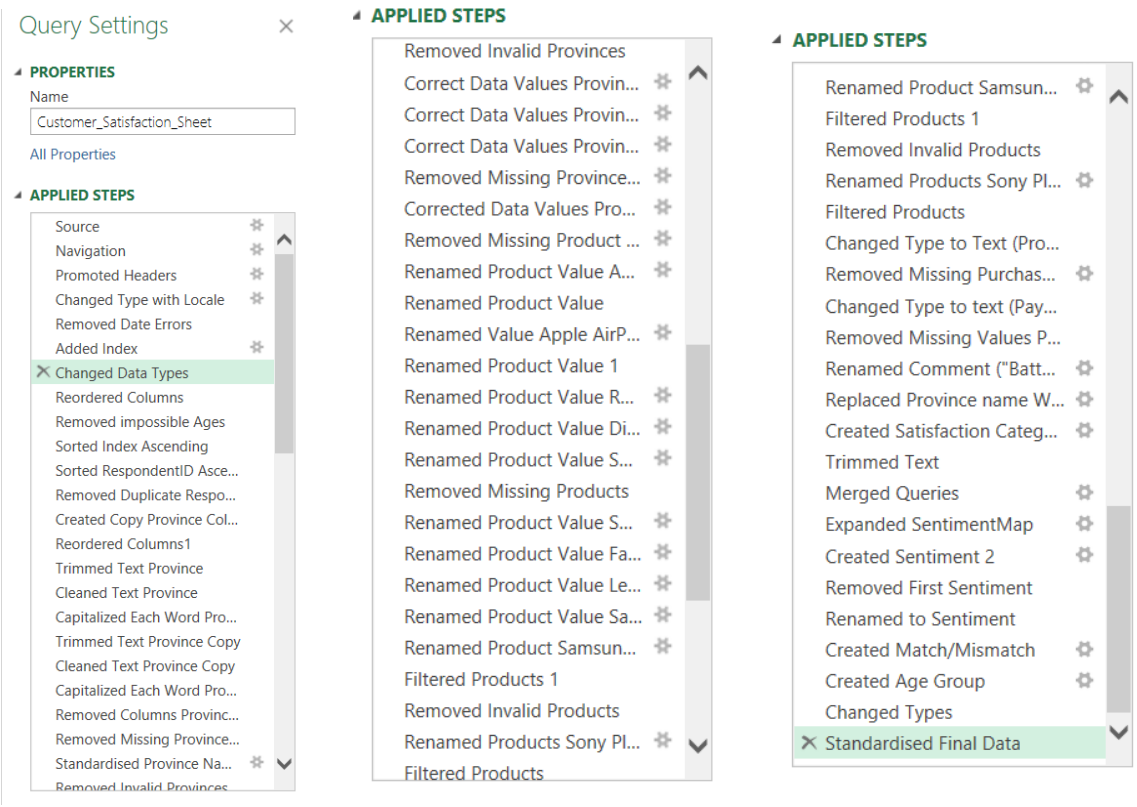


Figure 3: Power Query Applied Steps

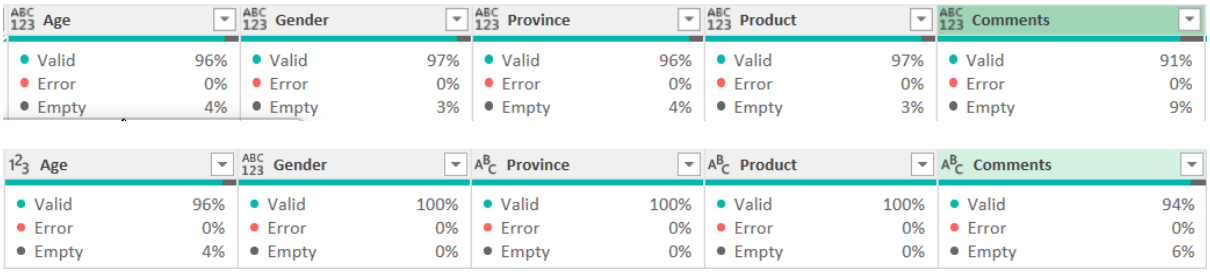


Figure 4: Before and after of Data Quality for specific columns

Step 7 — Created calculated/custom columns (Table: 3; Figure 5)

- Satisfaction Category
- Sentiment
- Rating–Comment Agreement (Match/Mismatch)
- Age Group

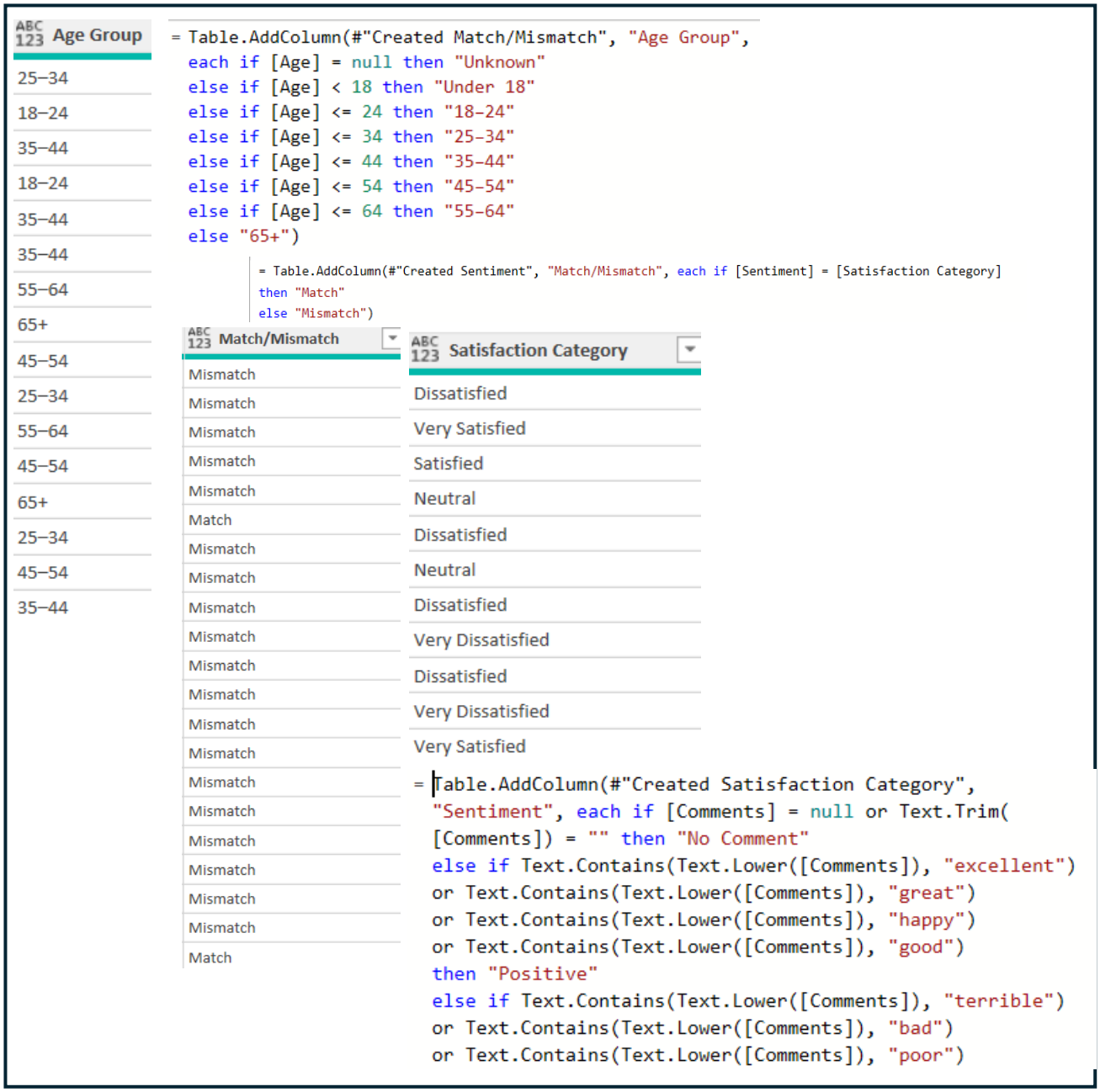


Figure 5: Custom Analytical Columns and Portions of their M Code

Table 3: New Columns and their Purposes

Created Columns	**Purpose**
Satisfaction Category	Categorises customer satisfaction
Sentiment	Categorises customer comments
Rating–Comment Agreement (Match/Mismatch)	Identifies whether satisfaction rating agrees with comment sentiment

Age-Group	Identifies different age ranges of respondents

6. Data Validation

Following transformation, the dataset was reviewed to confirm that data types were appropriate, categorical values were standardised, required fields were populated and calculated fields produced expected results.

That demonstrates that your cleaning process is repeatable, rather than simply showing the final cleaned data.

7. Final Dataset

After cleaning and transformation, the final dataset was loaded into Excel as the cleaned dataset (Figure used for PivotTable, PivotChart and dashboard analysis.

Final number of records: 8, 347

Final number of columns: 24

Important analytical fields: RespondentID, Satisfaction, Purchase Amount, Match/Mismatch, Product, Department, Province, Age Group, Gender, Recommend, Years (Date), Month (Date), Satisfaction Category, Sentiment.

Saved as an Excel Table: CleanedData.

Destination sheet: 01_Cleaned_Data in Customer_Satisfaction_Dashboard.xlsx

Index	RespondentID	Date	Age	Gender	Province	Product	Satisfaction	Recommend	Payment Method	Delivery Type	Loyalty Member	Comments	Satisfac
1	R00622	2025/03/31	55	Other	Mpumalanga	Laptop Backpack	2	No	Cash	Click & Collect	Yes	Very fast delivery	Dissatis
2	R04386	2025/09/04	34	Male	Mpumalanga	Facial Cleanser	5	Yes	Mobile Payment	Home Delivery	No	Arrived damaged	Very Sat
3	R06795	2024/04/11	18	Female	Eastern Cape	Skipping Rope	5	No	Debit Card	In Store	No	Easy to assemble	Very Sat
4	R04563	2024/12/20	60	Female	Limpopo	Laptop Backpack	3	Yes	EFT	In Store	Yes	Easy to assemble	Neutral
5	R09265	2024/04/20	57	Male	North West	Breakfast Cereal	5	No	Credit Card	Click & Collect	No	Exactly as advertised	Very Sat
6	R03133	2025/03/07	35	Male	Free State	Wireless Earbuds	1	No	Cash	Home Delivery	Yes	Would definitely buy again	Very Dis
7	R08711	2025/07/17	22	Other	Western Cape	Toy Train Set	1	Yes	Cash	Click & Collect	Yes	Packaging could improve	Very Dis
9	R07783	2024/09/21	60	Male	Mpumalanga	Toaster	1	Yes	Debit Card	Click & Collect	Yes	Excellent product	Very Dis
10	R06504	2024/09/06	51	Male	Eastern Cape	Logitech Mechanical Keyboard	2	No	Debit Card	Home Delivery	Yes	Great customer service	Dissatis
11	R00705	2024/08/30	72	Female	North West	Apple AirPods Pro	4	No	Credit Card	Click & Collect	Yes	Packaging could improve	Satisfie
12	R07667	2025/04/02	58	Female	Eastern Cape	Energy Drink	3	No	Cash	In Store	No	Instructions unclear	Neutral
13	R00097	2025/08/20	27	Male	Free State	Throw Pillow	3	No	Cash	Click & Collect	No	Easy to assemble	Neutral
14	R03119	2024/02/06	39	Male	Eastern Cape	Energy Drink	1	Yes	Credit Card	In Store	No	Great customer service	Very Dis
15	R02389	2024/08/13	28	Female	Eastern Cape	Sunscreen	3	No	Debit Card	Home Delivery	No	Item missing accessories	Neutral
16	R04813	2025/04/01	53	Female	Gauteng	Men's T-Shirt	5	Yes	EFT	Home Delivery	No	Packaging could improve	Very Sat
17	R02955	2024/11/26	71	Other	Limpopo	Coffee Machine	3	Yes	Credit Card	Click & Collect	Yes	Average quality	Neutral
18	R03528	2025/04/24	79	Other	Western Cape	Hair Straightener	3	Yes	Debit Card	Home Delivery	Yes	Easy to assemble	Neutral
19	R04023	2024/03/15	68	Other	Western Cape	Plush Teddy Bear	5	No	EFT	In Store	Yes	Arrived damaged	Very Sat
20	R00511	2024/02/21	61	Other	North West	Watch Strap	5	Yes	Debit Card	Click & Collect	Yes	Arrived damaged	Very Sat
21	R00305	2025/02/26	62	Female	Free State	Face Moisturizer	1	Yes	EFT	Home Delivery	Yes	Item missing accessories	Very Dis
22	R05897	2025/10/17	69	Other	Limpopo	Bookshelf	1	No	Credit Card	Click & Collect	Yes	Product exceeded expectations	Very Dis
23	R00034	2024/02/06	71	Female	Northern Cape	Organic Coffee Beans	4	Yes	Debit Card	Home Delivery	No	Product exceeded expectations	Satisfie
24	R08570	2025/11/09	47	Male	Free State	Winter Jacket	3	Yes	Mobile Payment	In Store	No	Average quality	Neutral

Figure 6: Final Cleaned Dataset

8. Data Transformation → Analysis Workflow

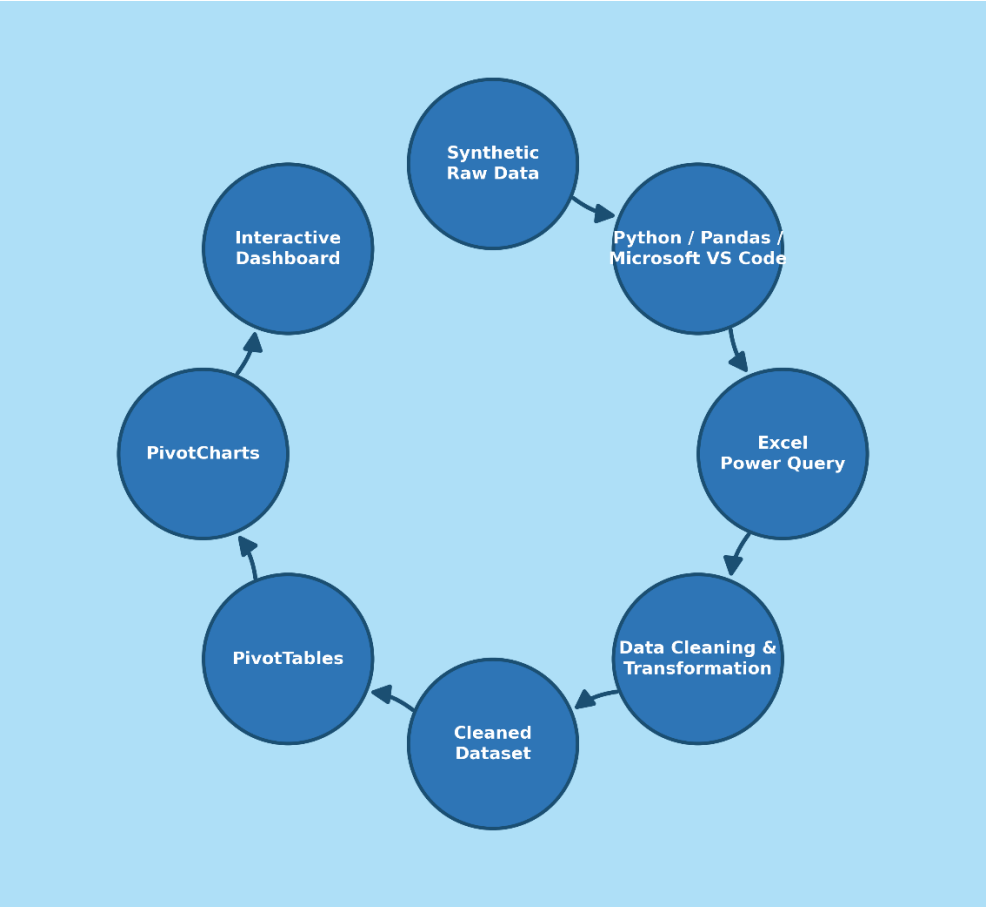


Figure 7: Flow Chart of the entire Data Transformation process from beginning to end

9. Tools Used

Table 4: Tools/Formulas and Features used in the Portfolio and their purposes

Tool/Formula/Feature	Purpose
Python / Pandas/VS Code	Synthetic data generation / initial data preparation
Excel Power Query	Data cleaning and transformation
IF	Create logical classifications or checks
XLOOKUP / VLOOKUP	Look up product, department, region information and purchase amounts from your lookup table
RANDBETWEEN	Generate synthetic/random purchase amounts during dataset preparation
GETPIVOTDATA	Pull KPI values from your PivotTables into the dashboard
TEXT / concatenation (&)	Format KPI displays such as 4.3 / 5
Shape Configuration	Aligning block shapes correctly to create KPI cards
Excel PivotTables	Aggregation and analysis
Excel PivotCharts	Data visualisation
Excel Slicers	Interactive filtering
Excel	Dashboard development

10. Conclusion

Power Query provided a repeatable process for cleaning and transforming the customer satisfaction dataset. The resulting structured dataset was used as the foundation for the PivotTable analysis and interactive Excel dashboards.

Appendix A: Detailed Data Cleaning steps column by column in Power Query for Retail Customer Satisfaction raw data

Primary: Created Index column and sorted in ascending by RespondentID and removed RespondentID duplicates

A. Date

Corrected 10068 date formatting issues using convert to date format and locale conversion (Figure 8).

Removed 82 impossible dates because the correct dates could not be verified.

B. Age

There were 381 null values (Figure 8). Filtered impossible values out (-5,0,8,130,180).

C. Gender

Standardized categories (Male, Female, Other)

D. Province

There were 376 null values. Filtered out null values. Trimmed and cleaned column. Standardized values to official province names by correcting spelling variations and inconsistent naming conventions. Examples included 'Western Cap' → 'Western Cape' and 'KZN' → ‘KWAZULU-NATAL’.

E. Product

There were 369 null values. Filtered out null values. Standardized values to official product names by correcting spelling variations and inconsistent naming conventions. Examples included ‘Apple Air Pod Pro’ → ‘Apple AirPods Pro’ and ‘Sony Playstation5’→ ‘Sony PlayStation5’. Samsung

F. Recommend

Standardized categories (Yes and No).

G. Purchase Amount

Values for Purchase amount were not accurate for realistic data as produced by the Python script. Therefor a second Python Script was produced to produce a realistic

price for each product. A further XLOOKUP function on Excel was produced and randomized using the RANDBETWEEN function for the finalized Purchased amount.

H. Payment Method

Corrected 10068 formatting issues. Converted Numeral format to text.

I. Delivery Type

Corrected 10068 formatting issues. Converted Numeral format to text.

J. Loyalty Member

Standardized categories (Yes and No). Converted Numeral format to text.

K. Comments

Kept 637 null values. It's normal for many respondents to leave comments blank. Converted Numeral format to text. Converted 'Battery life disappointing' to 'Product was disappointing' because some of the products described were not electronic.

L. Department

Created department.py to create the lookup table to do VLOOKUP to showcase the department for each product.

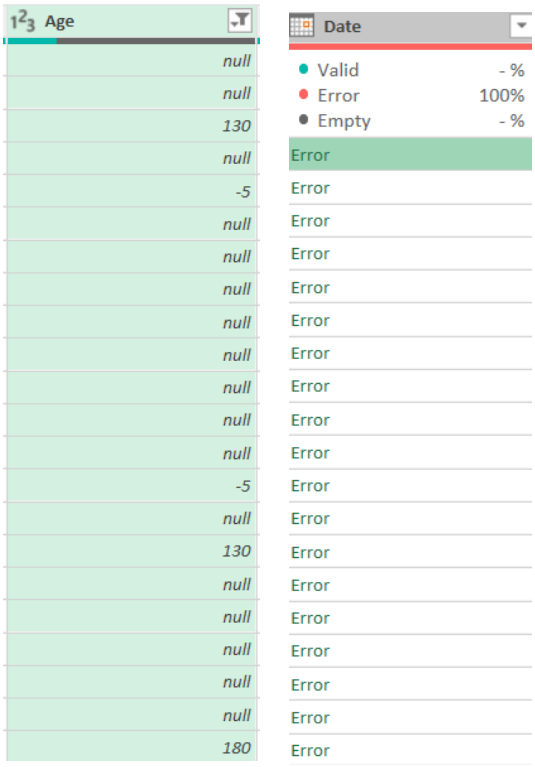


Figure 8: Data Type Validation (Columns showing the errors before they were corrected in the Age and Date Columns).